$$Y_i(\mathbf{q}) = C_i(\mathbf{q})\widetilde{V}\left(R_i^\top \begin{bmatrix} q_x \\ q_y \\ 0 \end{bmatrix}\right) e^{-2\pi i \mathbf{q}\cdot\mathbf{t}_i} + N_i(\mathbf{q})$$

Where a pose is defined by (Ri,ti)

Y_i(q)= $\mathcal{F}_2\{P_R V\}(\boldsymbol{q}) = \widehat{V}\left(R^T(q_x, q_y, 0)^T\right).$

So how exactly do we recover V as desired?

Property used here

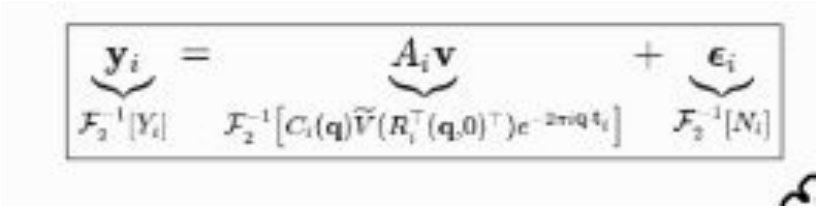
If $T_{\mathbf{t}}g(\mathbf{x}) = g(\mathbf{x} - \mathbf{t})$, then

$$\mathcal{F}[T_{\mathbf{t}}g](\mathbf{k}) = e^{-2\pi i \mathbf{k}\cdot\mathbf{t}}\widehat{g}(\mathbf{k}).$$

A translation becomes multiplication by a unit-magnitude complex phase. If $h * g$ is a spatial convolution,

$$\mathcal{F}[h * g](\mathbf{k}) = \widehat{h}(\mathbf{k})\widehat{g}(\mathbf{k}).$$

Reconstruction scheme 1 = you know the pose



First write previous equation in real space

First note that y_i is gaussian because the noise is gaussian

First, why Gaussian probability?

For one pixel, suppose the candidate volume predicts intensity μ, while the microscope measures y.

If the noise is Gaussian,

$$p(y \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y-\mu)^2}{2\sigma^2}\right].$$

For a vector of pixel observations, the corresponding multivariate Gaussian is

$$p(\mathbf{y}_i \mid \mathbf{v}) = \frac{1}{(2\pi)^{m/2}|\Sigma_i|^{1/2}} \exp\left[-\frac{1}{2}(\mathbf{y}_i - A_i\mathbf{v})^T \Sigma_i^{-1} (\mathbf{y}_i - A_i\mathbf{v})\right].$$

Here, $|\Sigma_i|$ is the matrix determinant, and Σ_i^{-1} is the *precision matrix* (the inverse covariance).

Now suppose there are N particle images.

Assuming conditional independence across particles, the joint likelihood is

$$p(\mathbf{y}_1, \dots, \mathbf{y}_N \mid \mathbf{v}) = \prod_{i=1}^{N} p(\mathbf{y}_i \mid \mathbf{v}).$$

Multiplying thousands of tiny Gaussian probabilities is mathematically inconvenient and numerically unstable.

Taking logarithms changes a product into a sum:

$$\log \prod_i p_i = \sum_i \log p_i.$$

For a vector of pixel observations, the corresponding multivariate Gaussian is

$$p(\mathbf{y}_i \mid \mathbf{v}) = \frac{1}{(2\pi)^{m/2}|\Sigma_i|^{1/2}} \exp\left[-\frac{1}{2}(\mathbf{y}_i - A_i\mathbf{v})^T\Sigma_i^{-1}(\mathbf{y}_i - A_i\mathbf{v})\right].$$

Here, $|\Sigma_i|$ is the matrix determinant, and Σ_i^{-1} is the *precision matrix* (the inverse covariance).

Take the log to get

$$L(\boldsymbol{v}) = \frac{1}{2}\sum_{i=1}^{N}(\boldsymbol{y}_i - A_i\boldsymbol{v})^T\Sigma_i^{-1}(\boldsymbol{y}_i - A_i\boldsymbol{v}). \tag{16}$$

Taking a gradient with respect to $\boldsymbol{v}$ gives

$$\nabla L = \sum_i A_i^T\Sigma_i^{-1}(A_i\boldsymbol{v} - \boldsymbol{y}_i).$$

Setting this equal to zero yields the normal equations:

$$\left(\sum_i A_i^T\Sigma_i^{-1}A_i\right)\widehat{\boldsymbol{v}} = \sum_i A_i^T\Sigma_i^{-1}\boldsymbol{y}_i. \tag{17}$$

Now \hat(v) can then be calculated